Cover Page

- **Problem Statement/Concept Title**

  AetherAI – AI Wearables for Respiratory Attacks

- **Theme**

  Healthcare

- **Team Name (Registered on portal)**

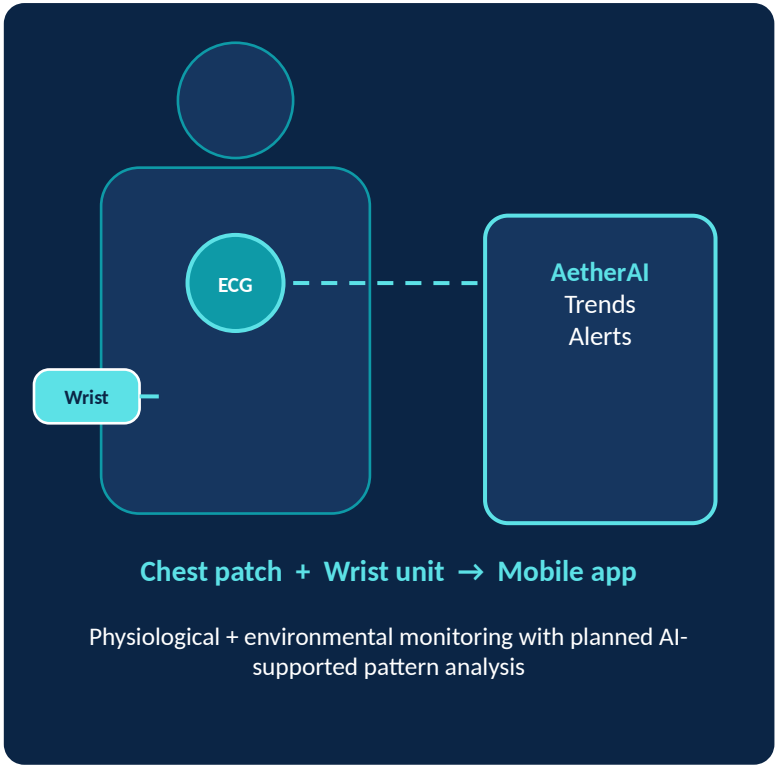
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- **Team ID (Registered on portal)**

  HSC|TN|02261

- **State/UT**

  Tamil Nadu



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Problem – Proposed Solution

Hack for Social Cause

Core problem

- Periodic measurements may miss changes between check-ups
- Tracking several measurements separately is inconvenient
- Environment gives useful respiratory context
- Users & caregivers need clear trends and timely notifications

Proposed solution

- Dual-unit wearable: chest patch + wrist unit
- Collects ECG, heart rate, SpO_2, motion & environment
- Sends readings wirelessly to a mobile app
- Explores AI analysis to flag unusual patterns that may warrant attention

Innovation & uniqueness

- Physiological + environmental data combined
- Separate chest and wrist sensing units
- Mobile visualization + planned ML analysis
- Potential for affordable, accessible respiratory-health awareness



AetherAI is a supplementary monitoring concept – it does not claim to diagnose or predict asthma attacks.

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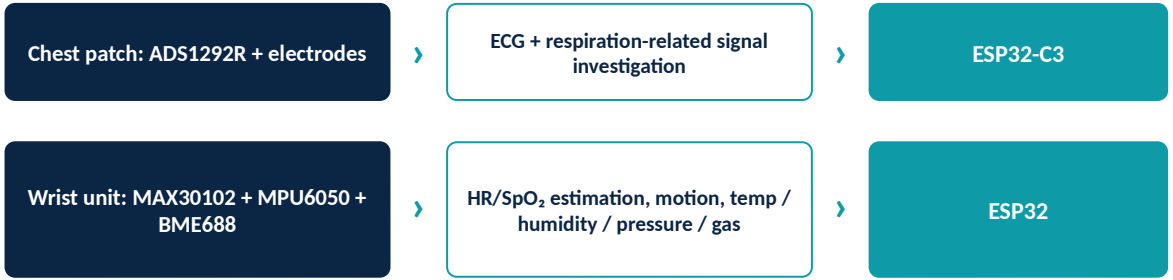
Technical Approach

Hack for Social Cause

Technologies to be used

- Controllers: ESP32, ESP32-C3
- Sensors: MAX30102, MPU6050, BME688, ADS1292R
- Wireless: Bluetooth / Wi-Fi (as suited)
- ML: Random Forest, Logistic Regression
- Mobile app: framework to be finalized (planned)

Methodology – hardware paths



Proposed system workflow (not every stage is implemented yet)



Repository & demo

- Repository: To be added
- Demo video: To be added

Extent of AI tool usage

- Project AI: Random Forest & Logistic Regression proposed for suitable labelled data
- Generative AI tools were used only to help prepare this presentation – separate from the project's ML

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Impact Potential

Hack for Social Cause

Target audience

- People living with asthma or COPD
- Caregivers and family members
- Clinics & healthcare institutions seeking supplementary monitoring

Potential benefits

- Convenient tracking of selected measurements
- Better visibility of health trends
- Potentially earlier awareness of unusual patterns
- Mobile notifications for configured conditions
- Supports remote-monitoring research & affordable wearable innovation

Stakeholders

- Users and caregivers
- Healthcare professionals
- Hospitals and clinics
- Educational & research institutions
- Healthcare technology developers

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UN SDG 3: Good Health and Well-Being

Secondary alignment: SDG 9 – Industry, Innovation and Infrastructure

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Feasibility and Risk Analysis

Hack for Social Cause

Feasibility

- Uses available embedded controllers & sensor modules
- Incremental build, starting with individual sensor tests
- Wireless link supports data transfer to a mobile app
- AI model evaluation on appropriately labelled datasets
- Prototype cost & battery life need component-level verification

#	Challenge / risk	Mitigation
1	**Sensor noise / unreliable readings**	Filtering, calibration, comparison with reference devices
2	**False alarms or missed patterns**	Independent model testing, threshold evaluation, careful alert design
3	**Battery limitations**	Low-power design; measure each module's consumption
4	**Wireless disconnection**	Connection-status monitoring; safe offline handling
5	**Health-data privacy**	Limited data collection, secure communication, access controls
6	**No clinical validation**	Supervised testing; consult qualified healthcare professionals before clinical use

AetherAI is a supplementary monitoring prototype, not a medical diagnostic device.

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Team

Team Composition

DG

Dev Guru Akhil Sekar

Software Developer

PS

Pazhanivelrajan S.

Hardware Developer

Roles

- Software: mobile app, data handling, ML analysis (planned)
- Hardware: sensor modules, ESP32 / ESP32-C3 integration

Challenges

- Sensor integration
- Signal quality
- Wireless communication
- Battery management
- Model validation

Journey

1. Prototype planning
2. Component integration
3. Mobile app development
4. Planned validation

Planned

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References

1. **Texas Instruments – ADS1292R product documentation**
   https://www.ti.com/product/ADS1292R
2. **Analog Devices – MAX30102 product documentation**
   https://www.analog.com/en/products/max30102.html
3. **Bosch Sensortec – BME688**
   https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors/bme688/
4. **TDK InvenSense – MPU-6050 product documentation**
   https://invensense.tdk.com/
5. **Espressif Systems – ESP32 documentation**
   https://docs.espressif.com/
6. **World Health Organization – Asthma**
   https://www.who.int/news-room/fact-sheets/detail/asthma
7. **World Health Organization – COPD**
   https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease
8. **Breiman, L. (2001). "Random Forests." Machine Learning, 45, 5–32.**
   https://doi.org/10.1023/A:1010933404324